

2024

The International Mathematical Modeling Challenge (IM²C) Summary Sheet

Developing Multi-scale LSTM Model for Stock Prediction

Summary Sheet

The earliest stock prediction method was weight-based machine learning, but as the data set increased, people realized that machine learning did not perform well on large data sets. To optimize machine learning, people turn to neural networks. Now, we designed a new way to predict stock by using multi-scale LSTM models and linear regression.

Our motivation for studying multi-scale LSTM models and linear regression stems from the pursuit of more accurate and comprehensive stock market predictions. The multi-scale LSTM model can synthesize information at different time scales, better capture market fluctuation characteristics, and improve prediction stability. The introduction of linear regression further improves the accuracy and stability of prediction by optimizing the combination of multi-scale information through learning weight distribution. By doing these studies, we can better cope with the limitations of single-scale models and avoid the challenges brought by market complexity and volatility, in order to provide more reliable and universal stock market prediction tools and provide investors with a more informed decision-making basis.

We use the Ford Motor Company stock data for our experiments as a dataset, including opening price, closing price, high price, low price, and trading volume. The data is divided into training groups, validation groups, and test groups, and 30, 45, and 60 days of data are used to predict the stock closing price of the next day by applying a multi-scale LSTM model to improve the accuracy of prediction. We designed a multi-scale LSTM model that combines LSTM models with three different scales (30, 45, 60) to improve the accuracy of stock predictions by calculating the average. Two error metrics are used to evaluate the model, mean square error (MSE) and mean absolute error (MAE). Experimental results demonstrate the predictive performance of three single-scale LSTM models (30, 45, 60) and merge them into a multi-scale LSTM model (LSTM-average). By observing the training loss and validation loss of each model, the trend of the model within 5000 epochs was analyzed. Experimental results show that LSTM-average is relatively good in terms of MAE and MSE.

Afterwards, in order to further improve the model, we added linear regression to form a new multi-scale LSTM model (LSTM-LR). By training the linear regression model, the weight parameters are obtained, and the predicted values of the three single-scale LSTM models are linearly combined. LSTM-LR performed well on the test set, significantly reduced the prediction error, and had better performance than other models. After comprehensively comparing the MAE and MSE of all models, it was found that LSTM-LR performed the best in terms of error.

Therefore, by adding linear regression and combining LSTM models of different scales, the prediction accuracy is successfully improved and provides an effective method for stock prediction. Our experiment shows that combining multi-scale LSTM models and linear regression methods can predict the stock market more accurately than using purely LSTM models. In multi-scale comparisons, the error values of some scales are not very different, and the combined multi-scale MAE and MSE are even larger than some single-scale values. Therefore, the introduction of linear regression further improves the accuracy of prediction, causing the MAE and MSE to be significantly reduced. Overall, we believe that combining multi-scale LSTM models and linear regression methods can better predict the stock market.

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1 Introduction

1.1 Introduction of the Stock Market

Stock, also known as equity, is a kind of securities representing the parts of ownership in a publicly held company. A stock market allows the exchange of shares among buyers and holders of the shares. Companies issuing shares to the stock markets to raise finance for their use. Holders of the shares can get benefits from the long-term growth of the companies or sell the stock for a price difference. Typically, those financial activities take place in several stock exchanges. In the essay, we will primarily focus on the stock price of Ford Motor Company(F), a well-known vehicle company, whose stocks are exchanged on the Nasdaq exchange.

The origin of stocks can be traced back to the 17th century in the Netherlands. At that time, the Dutch East India Company became the first company in history to issue stocks to the public. To raise funds for expeditions and trade, the company issued stocks for investors to purchase a portion of ownership of the company. At the end of the 17th century, the London Stock Exchange in England was established, marking the formal formation of the stock trading market. In the 19th century, the rise of railways and industrial revolution propelled the rapid development of the stock market. To expand and develop capital, companies have issued stocks one after another. The New York Stock Exchange (NYSE) in the United States was founded in 1792 and has become one of the largest stock exchanges in the world, witnessing the rise of the US stock market. However, the stock market crash at the beginning of the 20th century led to the Great Depression, and the stock market faced serious challenges. Subsequently, the formulation of securities regulations and the establishment of regulatory agencies became key to maintaining market order and protecting investors. The rapid development of information technology has driven the rise of electronic trading systems, making stock trading more convenient and efficient. Investors can easily trade in the stock markets of different countries.¹

The price of the stock is nonlinear, and there are numerous influence factors in the stock market, so it's hard to predict the stock price. The price of stocks is influenced by various factors such as supply and demand, company performance, expectations of the shareholders, etc. This makes the stock market full of volatility. Secondly, stocks have the characteristic of high liquidity, making it relatively easy for investors to buy and sell stocks. Investors can flexibly adjust their investment plans through the stock market to adapt to market changes.

On the other hand, investor behavior and market psychology also play a crucial role in the formation of stock prices. Market sentiment and the psychological state of investors have a non-linear impact on stock prices. For example, excessive market reactions and emotional fluctuations may lead to irrational price fluctuations that are inconsistent with market fundamentals. When the government's economic policies are adjusted, it will lead to the same nonlinear market model.²

¹ See Reference [1]

² See Reference [2]

Stock predictions have a lot of meaning for both investors and the financial industry. First, for investors, accurate stock price predictions help formulate smarter investment strategies. Investors can choose better buying and selling opportunities based on the prediction of future market trends, optimize their investment portfolio to the greatest extent, and achieve a risk-return balance. Second, stock price predictions help investors identify potential risks and take corresponding risk management measures to reduce portfolio volatility and improve overall asset management efficiency. Third, for listed companies, understanding the market's expectations for their stocks is crucial to making corporate strategies and financial decisions. Company management can adjust strategies based on market response and make decisions such as capital raising, and acquisitions.

1.2 Introduction of Candlestick Charts

Candlestick charts are graphs that represent financial price fluctuations over a certain period of time using rectangles. Each rectangle includes four key price data points, which are the highest price, the lowest price, the opening price, and the closing price.

1. The highest price: The highest point at the top of the rectangle represents the highest trading price of the asset during that time period.
2. The lowest price: The lowest point at the bottom of the rectangle represents the lowest trading price of the asset at a specific time.
3. Opening price: The starting point of the rectangle, represents the trading price at the beginning of a specific time period.
4. Closing price: The end point of the rectangle, represents the trading price at the end of a specific time period.

These four price points together form a complete rectangle, displaying the price trend of assets during a specific time period. The color of the rectangle is also an important source of information, usually green or white indicates a closing price higher than the opening price, while red or black indicates a closing price lower than the opening price.

1.3 The History of Candlestick Charts

Candlestick charts originated in 18th-century Japan, where the economy was prosperous, especially in the rice market. Farmers and merchants need a concise and effective way to understand the fluctuations in rice prices. In this context, Japanese merchants invented a table that records the prices and related information of different types of rice by plotting four price points for different time periods: opening price, closing price, highest price, and lowest price. This is the initial appearance of the candlestick chart. Later, in the 1980s, technical analysis became increasingly popular in financial markets, and candlestick charts became an important tool for investors and traders. After hundreds of years of evolution, the basic principle of candlestick charts has remained unchanged, but in the digital age, the production and analysis of charts have become more precise and efficient. Different charts represent different emotions and trends in the market, just like various combinations in candles. In the

figure, the red part in the blue box means when it is at 12:10 a.m., the opening price is 11.46 million dollars, and the closing price is 11.42 million dollars.

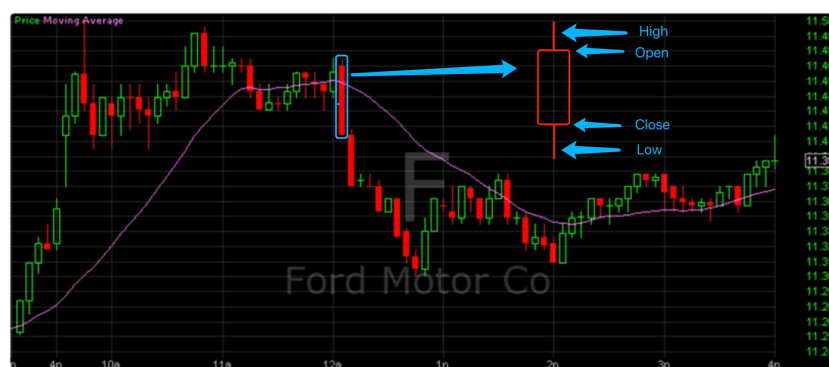


Figure 1: Daily Candlestick Chart of Ford Motor Company on Jan 25, 2024³

2 Proposed Methods

2.1 Basic concepts

2.1.1 Backgrounds

To have a better understanding of the principles of LSTM, we should understand how people predict stocks using previous methods. To begin with, machine learning is the earliest method of predicting stock, which can be traced back to 2006, using the weighting method⁴. However, when the dataset gradually increased, people realized that machine learning could no longer be a proper method to predict stock since machine learning can only be effective based on small datasets. To optimize machine learning, people optimized neural networks. A neural network is a type of machine learning algorithm loosely modeled on the structure and function of the human brain. Therefore, people need to develop different neural networks when facing different situations⁵.

2.1.2 Artificial Neural Networks (ANNs)

In recent years, ChatGPT, a chatbot developed by OpenAI, surprised many people all over the world, and Artificial Neural Network (ANN) is the lowest level logic. To be more specific, Artificial Neural Networks (ANNs) are computational processing systems that are mostly influenced by biological nervous systems, such as human brains. Also, Artificial Neural Networks (ANNs) are comprised of many interconnected computational nodes, which jointly learn from the input and compute the best result to output. For example, the human brain is a typical Artificial Neural Network (ANN), consisting of billions of neurons that are interconnected through synapses. The interconnectedness of neurons allows the brain to perform complex computations, learn from experience, and exhibit intelligent behavior, such as the instructions to squat, stand up, or walk.

³ See Reference [3]

⁴ See Reference [4]

⁵ See Reference [5]

ANNs (Artificial Neural Networks) are one of the simplest variants of neural networks. They pass information in one direction, through various input nodes, until it makes it to the output node. To be more specific, Artificial Neural Networks (ANNs) are computational processing systems that are mostly influenced by biological nervous systems, such as human brains. Also, Artificial Neural Networks (ANNs) are comprised of many interconnected computational nodes, which jointly learn from the input and compute the best result to output. For example, the human brain is a typical Artificial Neural Network (ANN), consisting of billions of neurons that are interconnected through synapses. The interconnectedness of neurons allows the brain to perform complex computations, learn from experience, and exhibit intelligent behavior, such as the instructions to squat, stand up, or walk.

ANN has 3 parts are input layer, a hidden layer, and an output layer. At the input layer, each node will receive different information, and then this information will be passed to the hidden layer with different weights, that is, the importance of the information. The hidden layers will then make decisions from the previous layer and weigh up how a stochastic change within itself detracts or improves the final output. The figure illustrates the 3 layers of ANN:

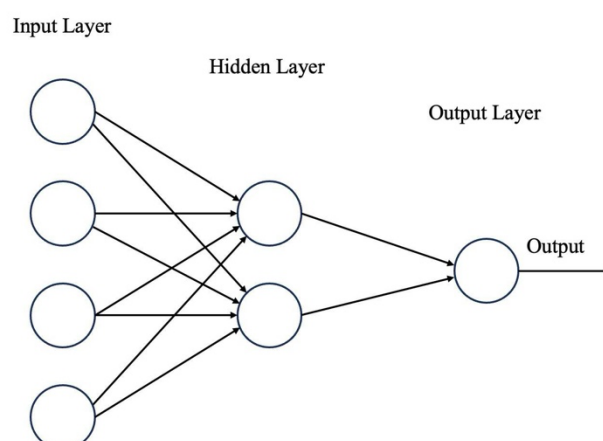


Figure 2: Basic Structure of ANN⁶

2.1.3 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) which are like ANNs are still comprised of neurons that self-optimize through learning. Although these two systems both use similar procedures, CNNs usually specialize in the image, using convolutional layers. To put it simply, for CNN, we input different pictures, let the system classify and process them by itself, and finally get an output. So, what matters now is the internal structure of CNN. CNNs are comprised of three types of layers. These are convolutional layers, pooling layers, and fully-connected layers.

The convolutional layer calculates the output of neurons by connecting them to local regions of the input. This is done through the computation of the dot product between the weights of the neurons and the corresponding region in the input volume.

⁶ See Reference [6]

The layers' parameters focus on the use of learnable kernels. The kernels used in convolutional layers are typically small but extend throughout the entire depth of the input. When the data passes through a convolutional layer, each filter convolves across the spatial dimensions of the input, resulting in the generation of a 2D activation map. Each kernel in the convolutional layer generates an activation map, and these maps are stacked along the depth dimension to create the complete output volume of the convolutional layer. For example, here is a 6*6 image with figures. Then, there exists a filter kernel that lays over some of the pixels of the input image depending on the dimensions of the Kernel size. After that, with a specific stride, in this example is 1, and the kernel moves with 1 stride. The numbers in the image and kernel must correspond one to one. The figure illustrates the basic structure of the convolutional layer:

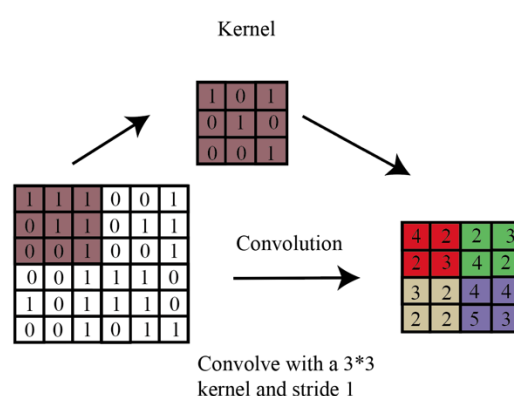


Figure 3: Basic Structure of the Convolutional Layer⁷

The pooling layer then conducts spatial downsampling of the input, effectively reducing the number of parameters within the corresponding activation. There are many kinds of pooling strategies, which may be the sum, the maximum number, the minimum number, and so on, and here is an example of the maximum number. The figure illustrates the basic structure of the pooling layer:

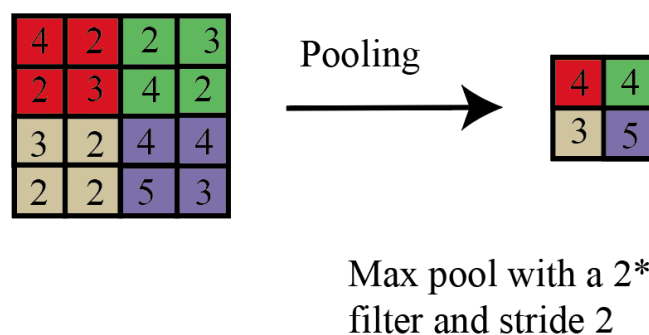


Figure 4: Basic Structure of the Pooling Layer⁸

⁷ See Reference [7]

⁸ See Reference [7]

The fully-connected layers, similar to standard ANNs, are responsible for generating class scores based on the activations. These class scores are then utilized for classification purposes. The figure illustrates the basic structure of the fully-connected layer.

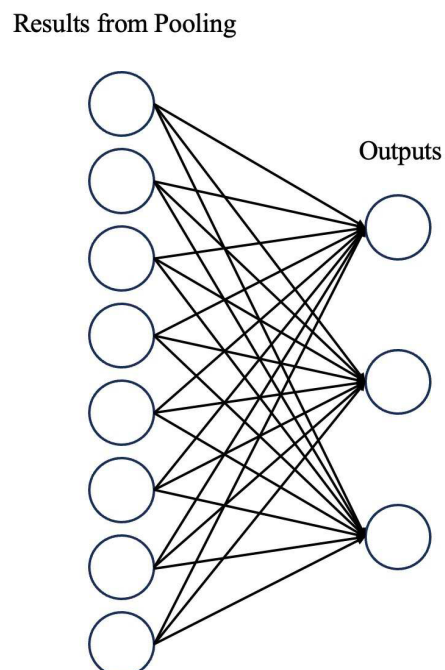


Figure 5: Basic Structure of the Fully-Connected Layer⁹

2.1.4 Recurrent neural networks (RNNs)

Recurrent neural networks (RNNs) are designed to interpret temporal or sequential information. RNNs can be achieved by accepting input and leveraging the activations of preceding or subsequent nodes within the sequence to affect the outcome. When working with temporal or sequential data, such as sentences, it is necessary to employ algorithms that can learn from both historical and future information in the sequence. The figure illustrates the basic structure of RNN.

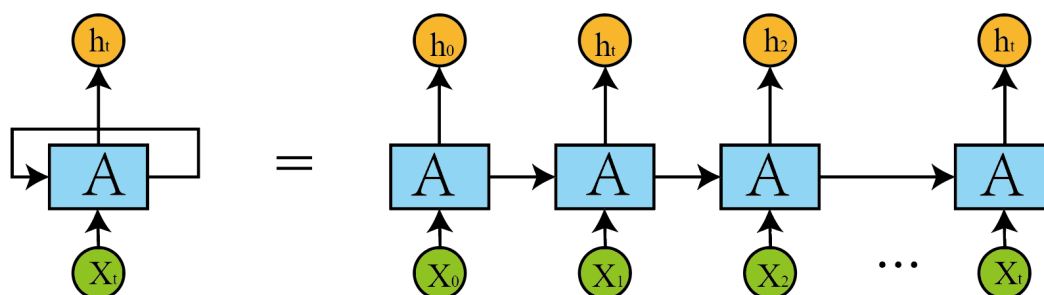


Figure 6: Basic Structure of RNN¹⁰

⁹ See Reference [6]

¹⁰ See Reference [6]

2.1.5 Deep Neural Networks (DNNs)

Deep Neural Networks (DNNs) are a subset of ANNs, which are also more complex than ANNs. They have more hidden layers and require handling large and high-dimensional datasets, but when training the datasets, DNNs are more challenging due to their depth¹¹.

2.1.6 Relationship with LSTM

Recurrent Neural Network (RNN) is one of the most used models in the field of deep learning, especially in dealing with time serial problems. Taking the information from the previous time slice as the current input, RNN is advantageous in treating the time series problem compared with traditional models such as Deep Neural Network (DNN). However, another problem in the deep learning field--Long Term Dependencies--is not solved in the RNN Model. Due to the vanishing gradient and exploding gradient problem in RNN, the model often loses the ability of long-term dependencies, losing its predictability in the next time slice. Long Short Term Memory (LSTM), as a special type of RNN model, is a solution to the long-term dependencies problem. By advancing the RNN model through ways like introducing gates to the learning process, LSTM performs even better in the prediction of information than the RNN model and contributes to different fields of life such as voice recognition and stock prediction.

2.2 LSTM Principles

LSTM (Long short-term memory) is a model built to solve the vanishing gradient problem in the RNN model. LSTM achieves "long short term memory" by continuously providing many short-term memories. LSTM models generally consist of the cell, input gate, output gate, and forget gate. Cells are used to store and transmit information. At each time step, the cell updates the content of the current cell based on the input, the memory cell content of the previous time step, and the control of the forget gate and input gate¹².

2.2.1 Input Gate

The input gate determines how much the value of the input node should be added to the internal state of the current storage unit. It uses a sigmoid activation function to generate an output between 0 and 1, which is used to determine how much information to retain.

$$i_t = \sigma \cdot (W_i \cdot [h_{t-1}, x_t] + b_i) \quad (1)$$

In the above formula, i_t means the output of the input gate; σ means the sigmoid activation function; W_i means the weight matrix of the input gate; $[h_{t-1}, x_t]$ means the concatenation of the hidden state h_{t-1} of the previous time step and the input x_t of the current time step; b_i means the bias of the input gate.

¹¹ See Reference [6]

¹² See Reference [8]

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (2)$$

The above formula, $\tilde{C}_t$ means unit status candidate value; $\tanh$ means the tanh activation function; W_c means the weight matrix connecting the hidden state h_{t-1} from the previous time step and the input x_t of the current time step; b_c means the bias of the candidate value.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (3)$$

In the above formula, C_t means updated cells; f_t means the output of the forget gate; C_{t-1} means the value of the memory cell at the previous time step $t - 1$.

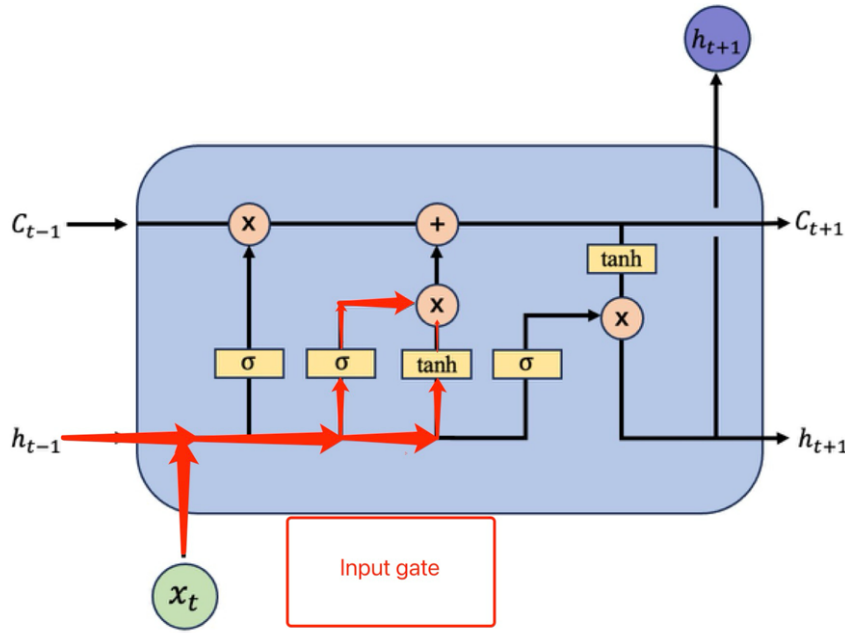


Figure 7: Steps of the Input Gate.¹³

2.2.2 Output Gate

The output gate determines whether the storage unit should affect the output of the current time step. It generates an output between 0 and 1 by using the sigmoid activation function and multiplies the output value and the content of the cell through the tanh activation function, thereby generating the hidden state for the current time step.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

In the above formula, o_t means the output of the output gate; W_o means the weight matrix of the output gate; b_o means the bias of the output gate.

$$h_t = o_t \cdot \tanh(C_t) \quad (5)$$

¹³ See Reference [9]

The above formula, h_t means the hidden state for the current time step.

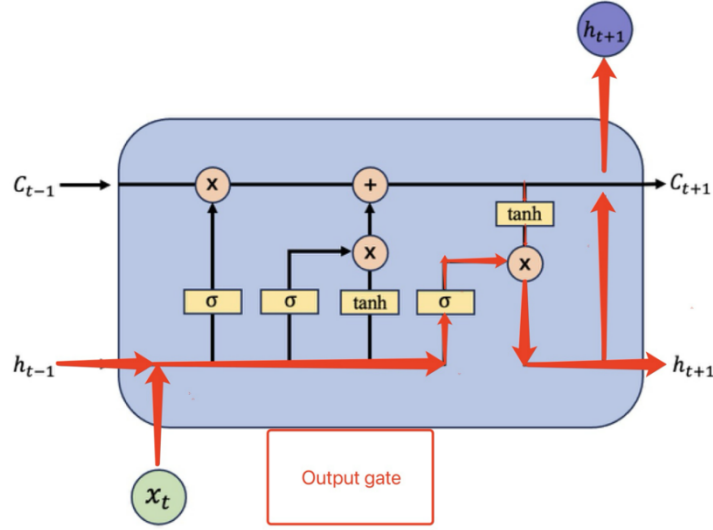


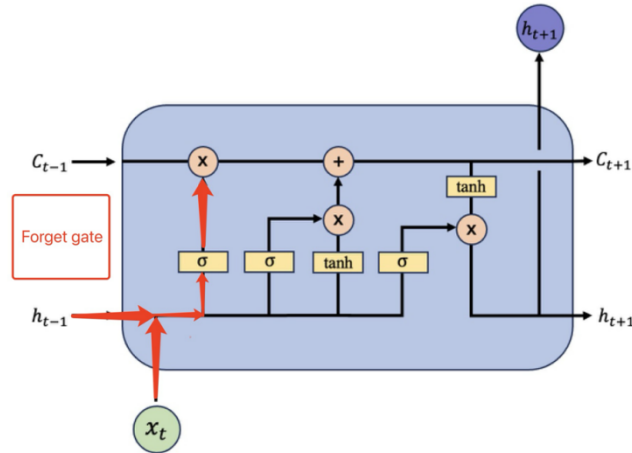
Figure 8: Steps of the Output Gate.¹⁴

2.2.3 Forget Gate

The forget gate obtains the appropriate forgetting ratio through learning. By using a sigmoid activation function to generate an output between 0 and 1, it determines what information is forgotten or retained.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (6)$$

In the above formula, W_f means the weight matrix of the forget gate; b_f means the bias of the forget gate.



¹⁴ See Reference [9]

Figure 9: Steps of the forget gate.¹⁵

2.3 Linear Regression Principle

Linear regression is introduced in our model. Least square method is used to fit data. The multiple linear regression model is:

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon \quad (7)$$

where p is the number of variables in the model.

For the evaluation of the model, the objective function is constructed:

$$J(\beta) = \sum_{i=1}^N (y'_i - y_i)^2 \quad (8)$$

where N is the number of data; y'_i is the real data; y_i is the value gotten from the model.

To minimize the objective function, the partial derivative can be used. In this way, β in the model can be found, and the model is formed.

3 Dataset

In this essay, we use Ford Motor Company (Ford), a publicly held vehicle company's stock, to do our experiment. The stock code of Ford is F. Ford did its initial public offering in 1956, at the NASDAQ stock exchange. The data has been collected from Yahoo Finance. We use the day chart price from 1977/1/3 to 2024/1/19. The data include the opening prices, the closing prices, the highest prices, the lowest prices, and the volume.

The data has been divided into three groups, the train group, the validation group, and the test group. The train group is the first 70% of the dataset, in other words, the day chart data from 1977/1/3 to 2010/7/22. The validation group contains the next 10% of the dataset, which means the data from 2010/7/23 to 2015/1/22. The test groups are the last 20% of data in the dataset, which is the data from 2015/1/23 to 2024/1/19.

By applying a multi-scale LSTM model to increase the accuracy. We use 30, 45, and 60 days of data to predict the next day's stock closing price.

4 Testing the Model

To improve the accuracy of the prediction of the stock by LSTM, we designed a multi-scale LSTM model, incorporating three LSTM models under different scales (30, 45, 60) by calculating the average. The incorporation is illustrated by the figure.

¹⁵ See Reference [9]

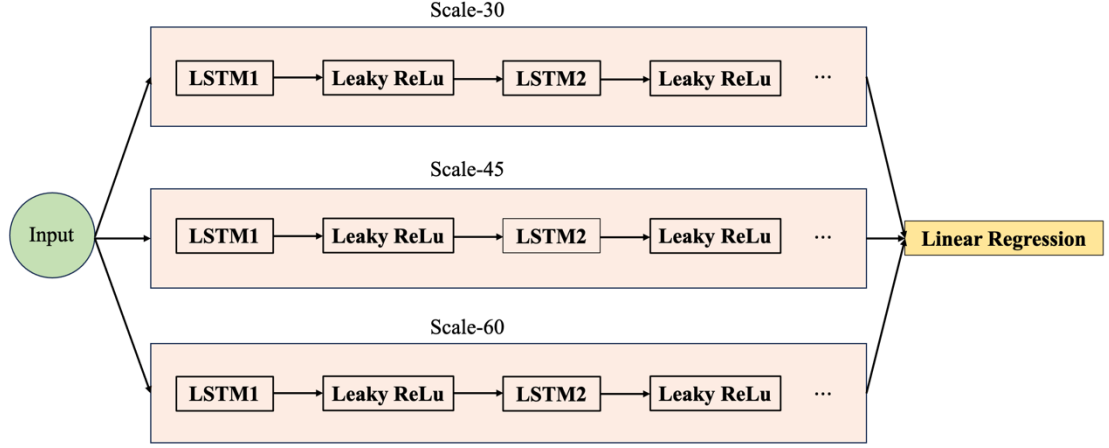


Figure 10: Structure of Multi-Scale LSTM Model

4.1 Basic Concepts

To evaluate our models, we use two error indicators: mean absolute error (MAE) and mean squared error (MSE).

4.1.1 Mean Absolute Error (MAE)

MAE indicates the absolute difference between the actual and predicted values, which makes it inherently robust to outliers. It reduces the effect of outliers on the loss function, dealing with errors equally. Under the circumstances in which there are outliers in the dataset, MAE is a better choice for error indicators than MSE. The formula for the calculation of MAE is:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y'_i - y_i| \quad (9)$$

In the formula, n is the number of pieces of values; y'_i is the predicted values in a specific date; y_i is the actual values in a specific date¹⁶.

4.1.2 Mean Squared Error (MSE)

MSE is a more widely used indicator for representation loss; it measures the average squared difference between actual and predicted values. In this way, the larger the errors are, the larger they impact the change of the indicator, as the difference is squared. The formula for the calculation of MSE is:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y'_i - y_i)^2 \quad (10)$$

In the formula, n is the number of pieces of values; y'_i is the predicted values in a specific date; y_i is the actual values in a specific date.

By adopting both indicators, we believe that the model can be evaluated in a comprehensive way¹⁷.

¹⁶ See Reference [10]

¹⁷ See Reference [10]

4.1.3 Train Loss and Validation Loss

The loss generated during the process of deep learning quantifies the errors made by our models. There are two types of loss: train loss and validation loss. Train loss measures the extent to which the model fits with the training data; validation loss measures the extent to which the model fits with the validation data. The purpose of paying attention to the data is to evaluate the performance of the model in each round and observe the trend and condition of the deep learning process¹⁸.

4.2 Experiments Results

4.2.1 Prediction by LSTM under a Scale of 30

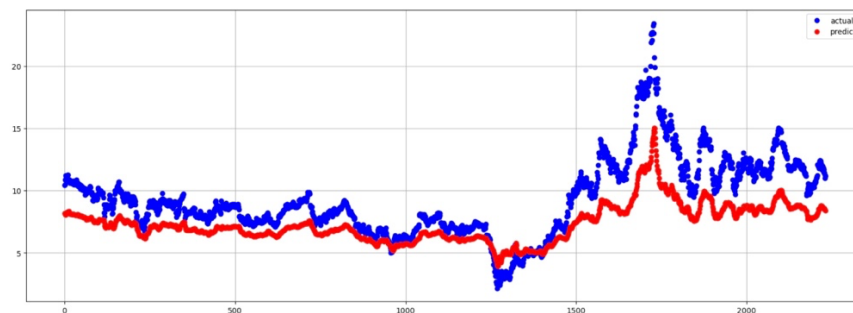


Figure 11: The Scatter Diagram of the Prediction by LSTM-30

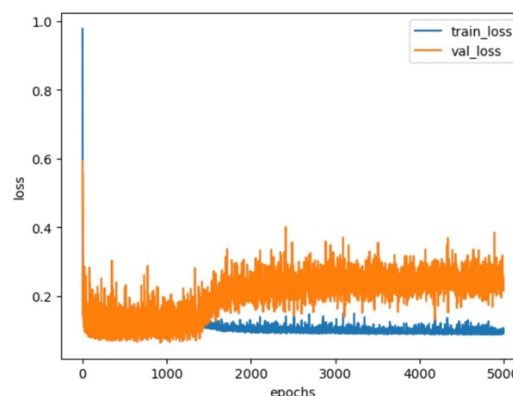


Figure 12: The Change of Losses during the Model Training Process

The scatter diagram for the LSTM model under a scale of 30 (LSTM-30) is presented in the figure [11]; the train loss and validation loss for each epoch are presented in the figure [12].

The model is trained for 5000 epochs, with 7855 pieces of data in the training dataset (from 1977/3/29 to 2010/7/23), and a decreasing (then slightly increasing for the validation loss) trend of losses can be seen in the figure [12]. At the beginning of the model training process (the first epoch), the train loss is 0.9784; the validation loss is 0.5916. In the end, the train loss decreases to 0.0915; the validation loss declines to 0.2374.

In the figure [11], the model predicts the close price of Ford Motor Company stock from 2015/4/22 to 2024/1/19. Through calculation, we get the MAE of the model to be

¹⁸ See Reference [11]

2.079; the MSE of the model is 6.504.

4.2.2 Prediction by LSTM under a Scale of 45

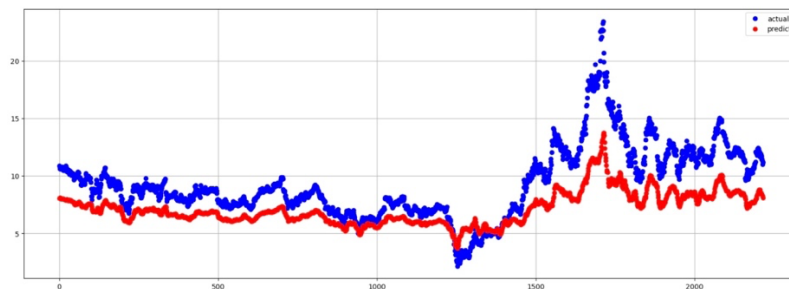


Figure 13: The Scatter Diagram of the Prediction by LSTM-45

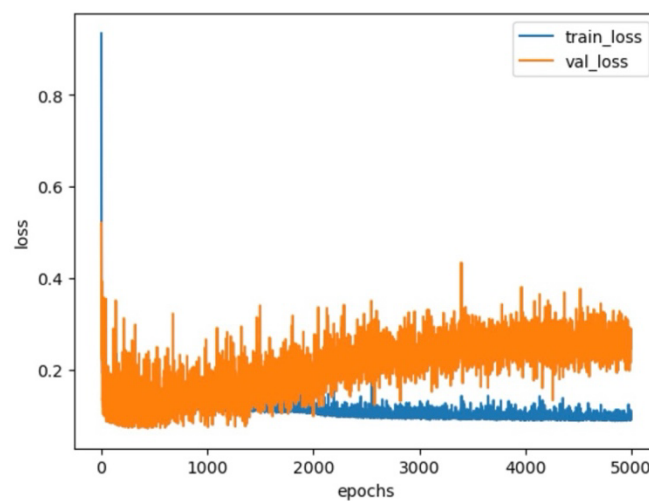


Figure 14: The Change of Losses during the Model Training process

The scatter diagram for the LSTM model under a scale of 45 (LSTM-45) is presented in the figure [13]; the train loss and validation loss for each epoch are presented in the figure [14].

The model is trained for 5000 epochs, with 7855 pieces of data in the training dataset (from 1977/3/29 to 2010/7/23), and a decreasing (then slightly increasing for the validation loss) trend of losses can be seen in the figure [14]. At the beginning of the model training process (the first epoch), the train loss is 0.9341; the validation loss is 0.5197. In the end, the train loss decreases to 0.1024; the validation loss declines to 0.2536.

In the figure [13], the model predicts the close price of Ford Motor Company stock from 2015/4/22 to 2024/1/19. Through calculation, we get the MAE of the model to be 2.301; the MSE of the model is 7.719.

4.2.3 Prediction by LSTM under a Scale of 60

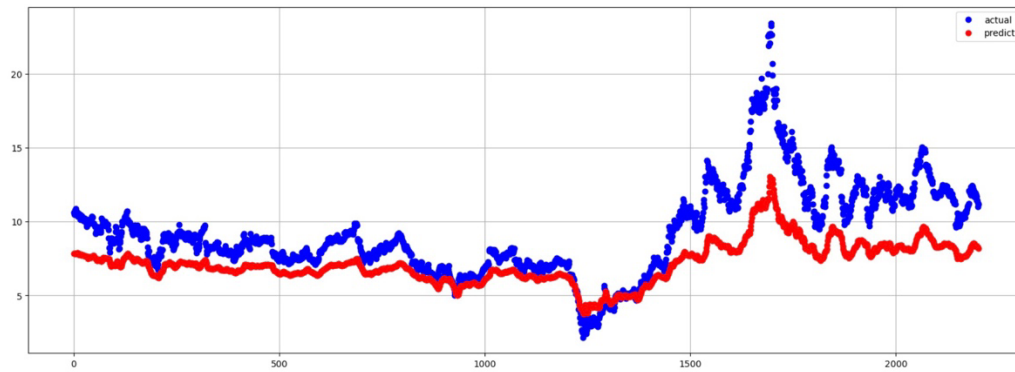


Figure 15: The Scatter Diagram of the Prediction by LSTM-60

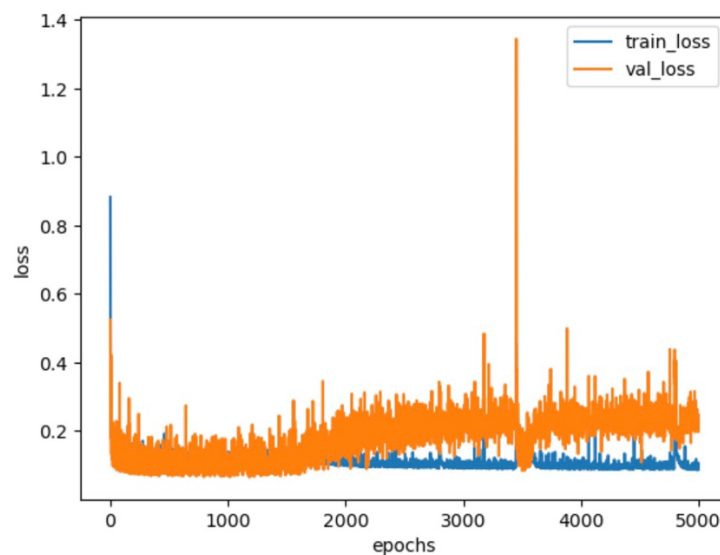


Figure 16: The Change of Losses during the Model Training Process

The scatter diagram for the LSTM model under a scale of 45 (LSTM-60) is presented in the figure [15]; the train loss and validation loss for each epoch are presented in the figure [16].

The model is trained for 5000 epochs, with 7855 pieces of data in the training dataset (from 1977/3/29 to 2010/7/23), and a decreasing (then slightly increasing for the validation loss) trend of losses can be seen in the figure [16]. At the beginning of the model training process (the first epoch), the train loss is 0.8821; the validation loss is 0.5241. In the end, the train loss decreases to 0.0883; the validation loss declines to 0.2455.

In the figure [15], the model predicts the close price of Ford Motor Company stock from 2015/4/22 to 2024/1/19. Through calculation, we get the MAE of the model to be 2.165; the MSE of the model is 7.516.

4.3 Incorporating Single-scale LSTM Models into Multi-scale LSTM

Model

The simplest way to combine the three single scale LSTM models we trained above is to calculate the average of the predicted values of the three models as the predicted value by the multi-scale LSTM model. In this way, the multi-scale LSTM model that combines three single-scale models by average is formed (LSTM-average). we can get the MAE to be 2.177; the MSE of the model is 7.219.

Model Name	MAE	MSE
LSTM-30	2.079	6.504
LSTM-45	2.301	7.719
LSTM-60	2.165	7.516
LSTM-average	2.177	7.219

Figure 17: The summary table of the errors of the four models

As it can be seen in the table, the multi-scale model LSTM-average performs better than LSTM-45 in terms of MAE and MSE and better than LSTM-60 in terms of MSE. However, the single-scale model LSTM-30 still performs greater than the multi-scale one; therefore, incorporating the single-scale models into the multi-scale one by calculating their average may not be an ideal way.

4.4 Application of Linear Regression

Hence, we introduced linear regression to form a new multi-scale LSTM model: LSTM-LR. We divided the 2202 pieces of data for the predicted value in the test dataset of each three single-scale models into two parts. The first part is the predicted close price from 2015/4/22 to 2019/4/9 (999 pieces); this part of the data is used for the training of the linear regression model, which gives us the weight of the predicted values from the three single-scale models that we need when calculating the predicted value by LSTM-LR. The rest pieces of the data (1203 pieces) were used for the testing and evaluation of the model. The formula of our linear regression model is:

$$Actual_{val} = \beta_0 + \beta_1 Predict_{val_{30}} + \beta_2 Predict_{val_{45}} + \beta_3 Predict_{val_{60}} + \varepsilon \quad (9)$$

In this model, $Actual_{val}$ is the real close price of the stock of Ford Motor Company from 2015/4/22 to 2019/4/9; $Predict_{val_{30}}$ is the predicted close price of the stock of Ford Motor Company from 2015/4/22 to 2019/4/9 by LSTM-30; $Predict_{val_{45}}$ is the predicted close price of the stock of Ford Motor Company from 2015/4/22 to 2019/4/9 by LSTM-45; $Predict_{val_{60}}$ is the predicted close price of the stock of Ford Motor Company from 2015/4/22 to 2019/4/9 by LSTM-60.

After the training of the linear regression model, we get β_0 as -3.793, β_1 as 2.122, β_2 as 0.363, β_3 -0.717. We then input the rest pieces of data, forming the predicted close price by LSTM-LR in the range from 2019/4/10 to 2024/1/19 (1203 pieces). The

prediction of close price is shown in the figure [18].

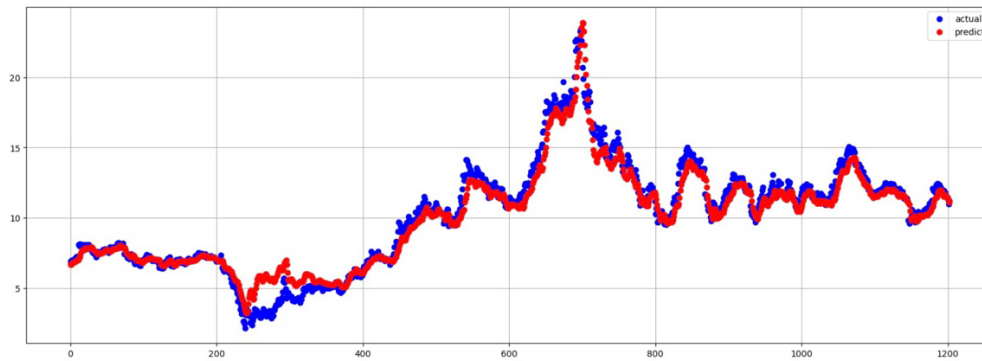


Figure 18: The Scatter Diagram of the Prediction by LSTM-LR

By comparing the actual and predicted values, we calculated the MAE and MSE of LSTM-LR. The MAE of the model is 0.6099; the MSE of the model is 0.7741.

Model Name	MAE	MSE
LSTM-30	2.079	6.504
LSTM-45	2.301	7.719
LSTM-60	2.165	7.516
LSTM-average	2.177	7.219
LSTM-LR	0.6099	0.7741

Figure 19: The Summary Table of the Errors of the Five Models

As shown by the table, by incorporating three single-scale LSTM models into LSTM-LR through linear regression, we successfully reduced the errors of the prediction significantly; LSTM-LR performs much better than all the previous models.

5 Strengths and Weaknesses

5.1 Strength

By introducing linear regression, the multi-scale LSTM model, LSTM-LR, predicts much more accurately compared with the LSTM-average and the other three single-scale LSTM models. It solves the problem of inaccuracy for the single-scale LSTM model in stock prediction due to factors like the random number in the model.

The model can be used to predict any type of price (close, open, volume...); it does not have strict requirements for the data, so it can be widely used.

The model can be used to predict stock in both short-term and long-term. By training the model with more data (price in seconds, minutes, or hours), the prediction will be even more accurate.

5.2 Weaknesses

In our model, we incorporate three single-scale models to form LSTM-LR. However, it is unknown whether incorporating more models (incorporating LSTM model under a scale of 75 or 90, for example) can form an even more effective model in the prediction of stock. The number of models may be determined in future research to generate a better multi-scale model.

In our research, each of the three single-scale models predicts 2202 pieces of data in the test dataset. We used 999 pieces of them for the training of linear regression and 1203 pieces for the evaluation of the model. We are not sure about how many pieces of data should be used for the training of the linear regression model to reach the best effectiveness of the prediction.

6 Conclusion

Using a pure LSTM model to predict the stock market is useful, but it can be more precise and accurate by using our proposed method: multi-scale LSTM model and linear regression.

At first, we weren't sure if our multi-scale LSTM model would be more accurate than purely using the LSTM model. We start by comparing the size of MAE and MSE at each scale with the size of MAE and MSE combined at several scales. We find that several values are not very different, and the combined MAE and MSE are even larger than the MAE and MSE of some scales. Faced with this situation, we used linear regression to make more accurate predictions. It turns out that more accurate predictions can indeed be made by linear regression because its MAE and MSE are much smaller than the MAE and MSE at all scales.

In conclusion, we believe that combining the multi-scale LSTM model and linear regression will have a better prediction of the stock market.

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